25 June 2024

Ref: CL/4475

Subject: **First consultation on the implementation of the 2021 Recommendation on Open Science**

Madam/Sir,

Under Article VIII of UNESCO's Constitution, Member States are required to submit regular reports on the measures they have taken to implement the conventions and recommendations adopted by the Organization.

In this context, I invite your Government to submit its report on the 2021 Recommendation on Open Science, in English or French, **no later than 28 February 2025**. Attached you will find guidelines to assist your authorities in submitting the report. Regional online sessions will also be organized to provide further guidance on report preparation.

Member States are encouraged to organize all necessary consultations with relevant ministries and institutions as well as with key stakeholders, including professional associations, civil society partners, private sector entities and National Commissions for UNESCO.

In addition, Member States are requested to designate a contact person responsible for information sharing and cooperation with UNESCO regarding reports on the 2021 Recommendation on Open Science. Member States should inform the Secretariat of this designation by 20 July 2024 at the latest at: openscience@unesco.org.

The Secretariat will present the first global report on Member States' application of this Recommendation for examination at the 222nd session of the Executive Board in autumn 2025. This report, together with the comments of the Committee on Conventions and Recommendations, will subsequently be submitted to the 43rd session of the General Conference in 2025.

The Natural Sciences Sector remains at your disposal for any further information you may require.

Please accept, Madam/Sir, the assurances of my highest consideration.

Audrey Azoulay
Director-General

Encl.: *Guidelines for the preparation of the reports from Member States on the implementation of the 2021 Recommendation on Open Science*

cc: Permanent Delegations to UNESCO
National Commissions for UNESCO
UNESCO regional offices
UNESCO field offices

To Ministers responsible for relations with UNESCO

7, place de Fontenoy
75352 Paris 07 SP, France
Tél. : +33 (0)1 45 68 10 00
Fax : +33 (0)1 45 68 55 55

www.unesco.org

ANNEX

GUIDELINES FOR THE PREPARATION OF THE REPORTS FROM MEMBER STATES ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

I. INTRODUCTION

These Guidelines are intended to assist Member States in the preparation of the reports on the implementation of the 2021 Recommendation on Open Science (hereinafter referred to as "the 2021 Recommendation") that was adopted by the 41st session of the General Conference of UNESCO on 23 November 2021.

The 2021 Recommendation proposes seven areas of action: (i) promoting a common understanding of open science, associated benefits and challenges, as well as diverse paths to open science; (ii) developing an enabling policy environment for open science; (iii) investing in open science infrastructures and services; (iv) investing in human resources, training, education, digital literacy and capacity building for open science; (v) fostering a culture of open science and aligning incentives for open science; (vi) promoting innovative approaches for open science at different stages of the scientific process; (vii) promoting international and multi-stakeholder cooperation in the context of open science and with view to reducing digital, technological and knowledge gaps.

For the purpose of the 2021 Recommendation, open science is defined as an inclusive construct that combines various movements and practices aiming to make multilingual scientific knowledge openly available, accessible and reusable for everyone, to increase scientific collaborations and sharing of information for the benefits of science and society, and to open the processes of scientific knowledge creation, evaluation and communication to societal actors beyond the traditional scientific community. It comprises all scientific disciplines and aspects of scholarly practices, including basic and applied sciences, natural and social sciences and the humanities, and it builds on the following key pillars: open scientific knowledge, open science infrastructures, science communication, open engagement of societal actors and open dialogue with other knowledge systems.

Pursuant to Articles 15 and 16.1 of the Rules of Procedure concerning recommendations to Member States and international conventions covered by the terms of Article IV, paragraph 4, of the UNESCO Constitution, the Director-General of UNESCO has invited Member States by the Circular Letter (CL/4381) to submit the 2021 Recommendation to their competent authorities within a period of one year from the close of the session of the General Conference at which it was adopted, i.e. before 24 November 2022.

Furthermore, under Article VIII of UNESCO's Constitution, Member States are required to submit a report on the action taken upon the conventions and recommendations adopted by the General Conference.

II. WHAT ARE THE AIMS OF THIS CONSULTATION?

This global consultation aims to assist Member States in: (1) mapping policies, mechanisms and actions related to open science and all its key pillars, against the objectives of this Recommendation; (2) collecting and disseminating progress and good practices on open science; (3) identifying challenges and opportunities faced by Member States in the implementation of the Recommendation, so as to identify specific capacity-building needs.

III. HOW TO FILL IN THE QUESTIONNAIRE?

The following questionnaire, which was developed with inputs from UNESCO working group on open science monitoring framework, aims to guide and assist Member States with their reporting on the progress made in the implementation of the 2021 Recommendation. It aims to collect information on the extent to which Member States have integrated the core values and guiding principles of open science that are defined in the 2021 Recommendation, in their national science, technology and innovation systems. Responses to this questionnaire will be considered as the official national report of each Member State.

Prior to completing the questionnaire, Member States are encouraged to organize the necessary consultations within and outside the concerned ministries and institutions, including with authorities and bodies responsible for science, technology and innovation, and consult relevant actors concerned with open science, including the scientific community, professional associations, civil society, indigenous and traditional knowledge holders, private sector partners and National Commissions for UNESCO.

In the preparation of their reports, Member States are also encouraged to consult the UNESCO Open Science Toolkit and the first edition of the *UNESCO Open Science Outlook* (published December 2023) containing a collection of information and communication materials relevant to the implementation of the 2021 Recommendation on Open Science. These resources are available online at https://www.unesco.org/en/open-science/toolkit and https://unesdoc.unesco.org/ark:/48223/pf0000387324.

Member States are requested to designate a contact person responsible for information sharing and cooperation with UNESCO in relation to reporting on the 2021 Recommendation.

Member States are encouraged to submit the questionnaire (in English or French) in one of the following ways:

(i) Online (preferred): the consolidated questionnaire (one per Member State) can be completed and submitted online.

(ii) Email: the questionnaire can be completed electronically and sent to: openscience@unesco.org

The responses to the questionnaire, if sent by email, should not exceed 15 pages, excluding annexes and is to be submitted to UNESCO in electronic form only (standard .doc, .pdf or .rtf format).

The reports will be made available on UNESCO's website in order to facilitate the exchange of information relating to the promotion and implementation of this Recommendation.

IV. DRAFT QUESTIONNAIRE

A. GENERAL INFORMATION ABOUT THE RESPONDANT:

- Country*: Hungary
- Organization(s) or entity(ies) responsible for the preparation of the report*:

  Name: Ministry of Culture and Innovation

  Website: https://kormany.hu/kulturalis-es-innovacios-miniszterium

  Email address*:

  Phone number*:

  Please describe the role/mandate of your organization: The Ministry of Culture and Innovation of Hungary is a governmental institution responsible for the fields of family and youth policy, culture, higher education, vocational and adult education, science policy, research and innovation.
- Officially designated contact person(s) who completed this survey:
  Full name*: Éva Bujtás
  Position*: science policy officier

Email address*:

- Full Name(s) of designated official(s) certifying the report: Ferenc Nagy-Rébék, Head of Department of Science Policy and International Scientific Cooperation

- Brief description of the consultation process established for the preparation of the report:

  Short consultation was proceeded with governmental institutions involved in open science

  and with the Hungarian National Commission for UNESCO.

- Other organization(s) or entity(ies) (including non-governmental) consulted for completing this survey:

  Name of the organization*: National Research, Development and Innovation Office

  Website*: https://nkfih.gov.hu

  Email address*

  Sector: ◯ Public ◯ Private ◯ Civil Society ◯ Other

B. Questionnaire for Member States' reporting on the implementation of the 2021 Recommendation

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

1.1 Has the 2021 Recommendation been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

- The former Ministry of Technology and Innovation shared the Recommendation 2021 with the appropiate Hungarian government agencies.
- The Secretariat of the Hungarian National Commission for UNESCO shared all information related to the draft recommendation and the final recommendation with the relevant ministry and, where applicable, with other relevant institutions.

YES / NO

If yes, please indicate which ministries and/or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

- Ministry of Foreign Affairs and Trade
- Hungarian Academy of Sciences
- Hungarian Research Network (former Eötvös Lorand Research Network)
- Hungarian National Commission for UNESCO

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

YES / NO

If yes, which language(s)? Hungarian
It was published on the website of the Hungarian National Commission for UNESCO in June 2024.

https://unesco.hu/data/Recommendation_on_Open_Science_hun_final.pdf

1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? *(ref.: (i) 16.f, 16.g, 16.h)*

YES / NO

If yes, please provide details on the activities organized or foreseen and links as relevant.

Position Paper on Open Science - National Research, Development and Innovation Office (NRDI Office)
https://nkfih.gov.hu/openscience/position-paper-on-open-science

Articles

On the website of the Hungarian National Commision for UNESCO,the following articles were published in order to promote open science in Hungary:

- Globális konzultáció a nyílt tudomány monitoringjáról
  (Global Consultation on the Draft Principles of Open Science Monitoring)
- UNESCO-ajánlás előkészítése a nyílt tudományról
- Ajánlást fogadott el az UNESCO a nyílt tudományról
  (UNESCO Recommendation on Open Science)
- Tagállami konzultáció a nyílt tudományról
  (Consultation of the states on open science)
- A nyílt tudomány és a koronavírus-járvány
  (The open science and the Covid-epidemic)

Events, conferences
Several events have been organized and will be held in the future in order to raise awareness on open science in Hungary. Here are some examples as follows:

- **Year of Hungarian Science 2025-2026**
  To celebrate the success of Hungarian science and Hungarian scientists, as well as the 200th anniversary of the foundation of the Hungarian Academy of Sciences and its Library, the Hungarian Parliament declared 2025 and 2026 the Year of Hungarian Science. (The 200th anniversary of the foundation of the Hungarian Academy of Sciences (1825) is on the list of anniversaries with which UNESCO is associated in 2024-2025.)
  The aim of the series of events is to present Hungarian science, research and innovation, as well as the past 200 years of the Hungarian Academy of Sciences, to emphasize the past and present social role of Hungarian science, research and innovation, the beauty, credibility and usefulness of science (social impact), as well as the economic impact of innovation results.
  The program of the Year of Hungarian Science will include events and experience-oriented science popularization programs aimed at the professional and general public.
  https://kormany.hu/hirek/20252026-a-magyar-tudomany-eve

- **NETWORKSHOP Conference 2025**
  13-15. May 2025
  The HUNGARNET Association will organize the 34th NETWORKSHOP conference at the Széchenyi István University in Győr. The key topics of the event are:
  - The digital transformation of higher education and public education, "Smart university";
  - General issues of data management, with special regard to the fate and management of FAIR research data;
  - Issues of information and network IT security;

  - Data repository systems and networks;
  - Data-driven decision-making: directions of data collection, how to process it and how to transform it into decision-supporting information;
  - The research methodology of digital humanities and the relationship between public collections and the network; Digital Humanities Platforms;
  - Trends in the application of AI (AI): service improvements in content delivery;
  - Cloud technologies; HUN-REN Cloud;
  - Digital infrastructures; Digital ecosystems;
  - Supercomputing (HPC);
  - GPGPU developments for AI purposes; International trends in research ICT infrastructures;
  - Research & Open Science WS; Collaboration platforms;
  - Smart learning environment; Learning Management Systems.

  https://nws.comp-rend.hu/

- **Hungarian Open Science Forum X.**
  - 28 November 2024
  - Experts and interested parties committed to open science met. The focus of the forum was on the training and development of open science professionals, with a particular focus on research data management, open access and the latest trends in open science.
  - https://openscience.hu/events/magyar-nyilt-tudomanyos-forum-x/

- **ARP Opening Ceremony**
  - 25 November 2024
  - The event featured presentations by the chief executives of the Hungarian Research Network (HUN-REN) and ARP creators on the importance of the project, the potential of the Data Repository Platform and related services. The presentations were followed by a round table discussion with representatives of domestic research institutions analyzing the situation of research data management in Hungary, and ARP Ambassadors reported on their experiences with the Data Repository and the reception of the services by fellow researchers. Finally, the new HUN-REN Data Steward Network was introduced.
  - https://researchdata.hu/en/esemenyek

- **Focus on Open Science Conference - 2024**
  - 21 November 2024
  - The Focus on Open Science Conference was organised by the Library and Information Centre of the Hungarian Academy of Sciences and Scientific Knowledge Services for the seventh time as a side event of the 2024 World Science Forum. The main theme of the event was Open Science and Citizen Science from different aspects. The aim of the event was to show how the European institutions are dealing with the changes related to the rise of open science.
  - https://eisz.mtak.hu/index.php/hu/rendezvenyek/232-focus-on-open-science-konferencia.html

- **NETWORKSHOP Conference 2024**

  3-5 April 2024

  The HUNGARNET Association organized the most prestigious Hungarian computer network and application IT conference, NETWORKSHOP, for the 33rd time. The workshop paid special attention to issues related to the development and application of artificial intelligence (AI). Other highlighted topics included cloud-based services, the use of quantum technologies, the use of various IT tools in research and the processing of public collection materials, as well as what else the digital future can bring. They presented the structure of the Data Repository Platform and the first lessons learned from its operation, and further described the developments of the data management profession.

- **Presentation on the use of HUN-REN ARP for users**
  20 March, 2024
  At the beginning of March 2024, HUN-REN opens the beta version of the ARP Data Repository to users. In order to enable the researchers and research support staff of HUN-REN to get to know the operation and services of ARP as much as possible, HUN-REN held online presentations for all those interested.

- **"Melody of Metadata" lecture**
  29 February 2024
  In the ever-evolving field of digital humanities, the role of reliable data management has become crucial. The presentation presented the management of the ARP Data Repository, as well as digital science strategies that can be used in the humanities, as well as the opportunities for cooperation.
  https://researchdata.hu/esemenyek/metaadatok-melodiaja

- **'Tripartite Event' about Open Science in Hungary – 2023**
  23 March 2023
  The National Research, Development and Innovation Office (NRDIO) hosted a 'Tripartite Event' supported by the European Commission (EC) and the European Open Science Cloud (EOSC) Association that provided a platform for participants to discuss the policies and best practices linked to Open Science. The aim of the event was to present the current position on Open Science and EOSC in Hungary and to exchange opinions with officers of the EOSC Association and the European Commission on the support policies and best practices in this field.
  At the hybrid event on 28 March 2023 the support for Open Science and FAIR data management were highlighted in the first panel discussion, which was attended by representatives of Hungarian research infrastructures: National Laboratory for Digital Heritage,ELI Alps, HUN-REN Cloud, OpenAIRE, OpenBioMaps.

- **Event series for the 20th anniversary of the Budapest Open Access Initiative - 2023**
  The Directory of Open Access Journals database has been helping the scientific community in its daily lives since 2003 by transparently sharing information about various open access journals and collecting the most useful data. In 2023, it celebrated its 20th anniversary with various events organized around three main themes: openness, globality, and trustworthiness.
  https://johncabot.libguides.com/blogs/LibraryNews/Budapest-Open-Access-Initiative-20th-Anniversary

- **National Open Science Forum**
  February 2023
  In 2021, the Governmental Agency for IT Development and the University and National Library of the University of Debrecen organized the first National Open Science Forum in order to deepen open science practices in the everyday work of the research community. In February 2023, the VI National Open Science Forum took place. Generally, the Forums are aimed at Hungarian researchers in Hungarian, but on 27 January, a Greek guest speaker shared his experiences with the audience in the European Open Science Cooperation (EOSC) in English.

  https://nkfih.gov.hu/english/events-of-the-office/nrdi-office-tripartite-event#:~:text=The%20National%20Research%2C%20Development%20and%20Innovation%20Office%20in,Hungary.%20Date%20and%20time%3A%2023%20March%202023.%2010%3A30-16%3A00

- **Domestic use of cutting-edge European IT research infrastructures within the framework of the SLICES ESFRI programme**

27 April 2023
A number of international state-of-the-art IT research infrastructures have become available to the scientific community – even with mobility support – within the framework of the SLICES programme, supported by the European Strategy Forum on Research Infrastructures (ESFRI), of which HUN-REN Cloud is also an active member. The event provided an insight into the 12 major European open research infrastructures of the SLICES programme, which effectively support the IT areas of machine learning, 5G/6G communication, cloud-edge computing and the Internet of Things (IoT), as well as their access and use methods for Hungarian researchers.
https://science-cloud.hu/en/events/opportunities-domestic-use-cutting-edge-european-it-research-infrastructures-under-slices

- **Open Science, Open Data in Hungary - Conference**
  16 November 2023
  The conference summared the first two years of the ARP in Hungary and was hosted by the HUN-REN Institute for Computer Science and Control (HUN-REN SZTAKI). During this event, attendees had the opportunity to explore presentations covering the historical background, current status, and future prospects of the HUN-REN Data Repository Platform (HUN-REN ARP).
  https://hun-ren.hu/research_news/open-science-open-data-in-hungary-conference-organised-by-hun-ren-arp-project-members-106778

- **NETWORKSHOP 2022 Conference**
  21 April 2022
  The HUNGARNET Association, in cooperation with the Ministry for Innovation and Technology and the Digital Success Programme organised the 31st edition of the NETWORKSHOP conference hosted by the University of Debrecen. On the second day of the event, 21 April, session was featured a series of presentations on the HUN-REN (former ELKH) Data Repository project.
  https://konferencia.unideb.hu/hu/networkshop2022

- **Connection of HUN-REN research infrastructures to the European Open Science Cloud (EOSC) programme**
  30 November 2023
  The aim of the programme was to familiarise Hungarian researchers with the European Open Science Cloud (EOSC) programme and to show how the Hungarian cloud-based infrastructure developments (HUN-REN Cloud and HUN-REN ARP) are connected to the EOSC programme. The knowledge presented will help Hungarian researchers to join the EOSC programme and to be able to use European infrastructures in addition to domestic infrastructures.

- **Data Economists in Hungarian Scientific Life - Event Series**
  23 November 2023
  In the field of research data management, professional organizations have formulated new principles and positions, researchers are facing previously unknown challenges, and institutions are working on new services. Since the autumn of 2023, institutionalized training in data economics has also been associated in Hungary. The task of the data economist is to help and guide researchers in the maze of data storage and sharing, and to support modern data management, keeping pace with technological innovations.

- **Open Science Lecture Series in Hungarian Higher Education**
  22-27 September 2022
  The Open Science Lecture Series in Hungarian Higher Education was held 22 times at 8 venues between 22 September and 9 December 2022. The topics of the lecture series included introduction to Open Science, the concept of open access, research data management, research evaluation systems, participants were introduced to the services of

the European Open Science Cooperation, and those who attended a workshop could gain knowledge about publications. The Government Information Technology Development Agency reached more than 370 students with a series of lectures.
https://openscience.hu/events/magyar-nyilt-tudomanyos-forum-vi/

1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[ii], in publicly funded research? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

YES / NO

If yes, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

National Declaration on Open Science of National Research, Development and Innovation Office (NRDI Office)
Hungary actively participates in and supports regional and international information technology infrastructures for open science. The National Research, Development and Innovation (NRDI) Office has been instrumental in promoting open science initiatives. In March 2021, the NRDI Office established the Open Science Advisory Board, which developed a National Position Paper on Open Science to guide the development of the open science ecosystem in Hungary.

The statement has been published in 2021 with the aim of expressing a common position on Open Science, based on professional consensus, in response to the current paradigm shift in the world of science, which, in line with the recommendations of international science organisations and the relevant policy objectives of the European Union, summarises the principles and the fields of activity of Open Science that best serve the interests and development of Hungarian science. The statement also aims to draw the attention of the scientific community in Hungary towards the importance and timeliness of the new approach, its strategic issues and the increasingly important role of Open Science in international cooperation.
https://nkfih.gov.hu/openscience/position-paper-on-open-science

Joint Declaration of the Presidents of National Academies of Sciences of the European Union Member States on robust, open and free science and education
A joint declaration has been issued in 2024 by the academies of science or equivalent scientific bodies of the 27 EU Member States, drawing attention to the importance of science and research. The declaration is addressed to all of the candidates standing for the European Parliament elections and, more broadly, to all interested parties, reminding them of the key role that science plays in the future of the European Union. The document was signed by Tamás Freund, President of the Hungarian Academy of Sciences, on behalf of the Hungarian Academy of Sciences.
https://easac.eu/fileadmin/user_upload/eu_statement_final_2.pdf

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so?

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

2.1 Does your country have a national policy[iii], strategy or plan of action on science, technology and innovation (STI)?

YES / NO

- **If yes** does it include any of the following open science elements:

- **open access to scientific knowledge**

The Open Access Initiative is a program, which provides free access to scientific documents. It's goal is to have peer-reviewed articles, which were published in journals that provide open access publishing, available freely to anyone and anywhere on the world. The self-archiving ,the uploading to the Instituitions's repositories can be done through the MTMT (Hungarian Scientific Publication Database) database.

- **open science infrastructure**

https://openscience.hu/hazai/

- **open engagement of societal actors**

- **Research, Development and Innovation Strategy of Hungary (2021-2030)** https://nkfih.gov.hu/english/research-development-innovation-strategy/proposal-of-the-rdi-strategy-of-hungary

- **National Declaration on Open Science** https://nkfih.gov.hu/nyilt-tudomany

- **Hungarian Act on Copyright (Act LXXVI of 1999)** https://net.jogtar.hu/jogszabaly?docid=99900076.tv

- **Amendments to Promote Open Access**
  - Act LXXVI of 1999 on copyright has undergone several amendments to address issues related to open access, particularly for research publications. These amendments align Hungarian copyright law with European Union directives on the Directive on Copyright in the Digital Single Market (EU 2019/790), which aims to increase the availability of scientific research and publications.
  - The Act includes provisions that encourage the open access publication of research articles and the use of publicly funded research results. It allows authors to retain certain rights over their work, enabling them to deposit their research in open access repositories without financial or legal barriers.

- **Hungarian National Data Archive (MTA SZTAKI)**
  - The Hungarian Academy of Sciences (MTA) – Institute for Computer Science and Control (SZTAKI) manages the Hungarian National Data Archive (HDA), which is an open access digital repository for research data. The HDA is part of Hungary's effort to provide a platform for storing, sharing, and disseminating scientific data, including datasets generated by publicly funded research.
  - The HDA ensures that scientific data is accessible, reusable, and preserved for future research. It encourages researchers to deposit their data in the repository, following open data principles to make the research process more transparent and reproducible.

- **Act CXII of 2011 on the Right to Informational Self-Determination and on the Freedom of Information**
  - The Act establishes principles of transparency and public access to government-held data in Hungary. It requires government agencies to publish certain types of data, including research data funded by public resources, in accessible formats.
  - Under this Act, public institutions, including universities and research organizations, are encouraged to make their research data open and accessible. This law aligns Hungary with EU open data policies and contributes to greater transparency in public sector research.
    https://njt.hu/jogszabaly/2011-112-00-00.2

- **Hungarian Implementation of the EU Open Data Directive**
  - Hungary is required to comply with the EU Open Data Directive (2019/1024), which mandates the open publication of government data and encourages public institutions to make research and scientific data publicly accessible. This Directive builds on the earlier Public Sector Information Directive (2003/98/EC), which introduced the legal framework for making public sector data available in reusable formats.
  - The Hungarian government has adapted its policies and laws to meet these requirements, ensuring that publicly funded research data and publications are accessible through national and institutional open access repositories.

- **European Open Science Cloud (EOSC) Participation**
  - The EOSC facilitates cross-border data sharing and promotes the use of open data repositories to make scientific research more transparent and collaborative. It is very important for the enhanced capacity HUN-REN Cloud to integrate into the European EOSC (European Open Science Cloud) program and the ESFRI infrastructure programs. To this end, HUN-REN SZTAKI is already participating in two European projects. In the EGI-ACE project, we ensure the compatibility of the HUN-REN Cloud with the EOSC clouds and provide user support for European projects where Hungarian consortium members are involved. In the SLICES-SC project, we are contributing to the preparation of the SLICES ESFRI program and organizing the European use of the SLICES infrastructure. Our goal is to participate in as many European projects as possible in the future, catering to the cloud requirements of the respective projects
  - Through its participation in EOSC, Hungary is contributing to the development of national and international infrastructures for data sharing, and Hungarian research institutions and researchers can take part in open science practices facilitated by EOSC.
    https://hun-ren.hu/open_science/hun-ren-cloud-107543

- **National Data Strategy for Open Science (2021)**

  Hungary has developed the National Data Strategy for Open Science (2021), which focuses on enhancing the availability of research data and ensuring that data produced through publicly funded research is accessible and reusable. This strategy aligns with the EU's open data agenda and focuses on creating a national infrastructure for open data, including open access digital repositories.

  The strategy also addresses issues such as data quality, metadata standards, and long-term data preservation, with a focus on enhancing collaboration among national research institutions and ensuring that data is freely accessible to all stakeholders. OpenScience_allasfoglalas.pdf

- **The Hungarian Academy of Sciences (MTA) and Open Access Initiatives**
  - The **Hungarian Academy of Sciences (MTA)** is actively involved in promoting open access through its **open access journals** and **repository for Hungarian research publications**. MTA's support for open access ensures that Hungarian researchers are provided with a platform to publish their research freely, while also complying with open access standards.
  - MTA also encourages the Hungarian research community to participate in international open access initiatives, contributing to the global movement for freely available scientific knowledge.

  https://openaccess.mtak.hu/wp-content/uploads/mta_hatarozat_oa_2016.pdf

  - **open dialogue with other knowledge systems**

  Directory of Open Access Journals (DOAJ)
  http://doaj.org/

Semmelweis University journals catalog
http://lib.semmelweis.hu/nav/folyoirat_katalogus

Budapest Open Access Initiative (BOAJ)
http://www.budapestopenaccessinitiative.org/read

Hungarian Open Access journals

Open Archives Initiative
http://www.openarchives.org

Registry of Open Access Repositories Mandatory Archiving Policies (ROARMAP) http://roarmap.eprints.org/

Berlin Declaration
https://openaccess.mpg.de/Berlin-Declaration

Please attach/upload a machine-readable PDF file of the policy and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[iv] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

- **National Declaration on Open Science**
  https://nkfih.gov.hu/nyilt-tudomany

- **Joint Declaration of the Presidents of National Academies of Sciences of the European Union Member States on robust, open and free science and education**

  A joint declaration has been issued on national level in 2024 by the academies of science or equivalent scientific bodies of the 27 EU Member States, drawing attention to the importance of science and research.
  https://easac.eu/fileadmin/user_upload/eu_statement_final_2.pdf

YES / PARTLY /NO

If yes or partly, please attach/upload machine-readable PDF file(s) of the policy(ies) and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policies.

2.3 In your country, are there specific policy instruments[v] that aim to promote open science?

YES/ UNDER DEVELOPMENT/NO

If yes or under development, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

Hungary has implemented specific policy instruments to promote open science. These efforts demonstrate Hungary's dedication to fostering an open and collaborative scientific environment through specific policy instruments and initiatives.

- In 2021, the National Research, Development and Innovation Office (NRDI Office) introduced the country's first national-level **National Declaration on Open Science**. This initiative aims to enhance transparency and collaboration in scientific research, making research, development, and innovation outcomes accessible to all. The Position Paper aligns with the European Union's policy objectives and reflects Hungary's commitment to the international open science movement.
  nkfih.gov.hu

- **Budapest Open Access Initiative**
  Hungary has been an active participant in the open access movement since the early 2000s. The Budapest Open Access Initiative, launched in 2002, is a significant milestone that advocates for free and unrestricted online access to scholarly research. This initiative has played a crucial role in shaping global open access policies and continues to influence Hungary's approach to open science.

- Hungarian academic institutions have established digital repositories to facilitate open access to scholarly works. The **HUNOR (HUNgarian Open Access Repositories**) consortium was established in 2008 by the libraries of Hungarian higher education institutions and the Library of the Hungarian Academy of Sciences to advance national open access practices and promoting the development and coordination of open access repositories across the country.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? *(ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)*

While a formal national strategy directly referencing the UNESCO Recommendation has not been published, Hungary's ongoing efforts in promoting open science principles indicate a positive trajectory towards embracing open science practices.

In 2021, the National Declaration on Open Science was joined by government and science-related umbrella organizations represented on the Advisory Board. Later on many Hungarian supporting organisation joined the Declaration initiated by the Hungarian Research, Development and Innovation Office. The Declaration contains the main pillars of the ecosystem on open science:

- ensuring open access to research results
- FAIR(1) and CARE(2) research data management,
- research integrity, scientific autonomy
- development of a new generation science evaluation system
- creation of new types of incentives
- international cooperation networks
- support community science
- education and training

YES / UNDER DEVELOPMENT /**NO**

If yes or under development, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? *(ref.: (ii) 17.d, 17.e, 17.g)*

YES /UNDER DEVELOPMENT/ NO

If yes, please provide more information.

The National Research, Development and Innovation Office (NRDI Office) has been actively promoting open science principles. In October 2021, the NRDI Office introduced Hungary's first national-level **Position Paper on Open Science**, aiming to enhance transparency and collaboration in scientific research. This initiative aligns with the European Union's policy objectives and reflects Hungary's commitment to the international open science movement.

2.6 In your country, are there specific funding mechanisms[vi] for open science at the national level? *(ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)*

YES / UNDER DEVELOPMENT / NO

If yes or under development, please provide details and links as relevant.

Hungary has demonstrated a commitment to open science through strategic initiatives and publisher agreements, dedicated national funding mechanisms are still evolving. Researchers are encouraged to explore both national strategies and European funding opportunities to support their open science endeavors.

However, at the present Hungarian researchers have opportunities to secure funding from European organizations that support open science. The European Research Council (ERC) offers grants for frontier research, emphasizing excellence and open access to scientific publications. Furthermore, the European Institute of Innovation and Technology (EIT), headquartered in Budapest, funds innovation and technology initiatives across Europe, potentially benefiting Hungarian researchers engaged in open science projects.

The National Declaration on Open Science, initiated by the NRDI Office in 2021, aims to address these issues by promoting open research practices and fostering collaboration among governmental and scientific institutions (Horizont 2020+; HUN-REN Cloud)

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such funding mechanisms.

2.7 In your country, have open science practices[vii] been integrated in existing research funding mechanisms?

YES /NO

Open science practices have been integrated into research funding mechanisms in Hungary. In October 2021, the National Research, Development and Innovation (NRDI) Office launched

the country's first N**ational Declaration Position Paper on Open Science**, signaling a commitment to making research, development, and innovation results accessible to all.
nkfih.gov.hu

Hungarian institutions have established agreements with publishers to support open access. For instance, there are **transformative agreements** in place that allow authors affiliated with Hungarian institutions to publish their research open access with Springer Nature's hybrid journals.
springernature.com

Furthermore, the NRDI Office offers funding schemes, such as the **Research Grant Hungary** (RGH_24), which support both basic and applied research projects. The Research Grant Hungary is a new funding scheme (2024) that provides priority support for up to five years to outstanding researchers in the most dynamic creative phase, at the forefront of international science, who are foreign nationals or live and work abroad.
https://nkfih.gov.hu/english/nrdi-fund/research-grant-hungary-rgh24/call-for-applications#:~:text=The%20Research%20Grant%20Hungary%20(RGH_24,in%20their%20field%2C%20contribute%20to

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? *(ref.: (ii) 17.i)*

YES / NO

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Hungary has demonstrated a commitment to fostering equitable public-private partnerships in open science, aligning with the 2021 UNESCO Recommendation on Open Science. In October 2021, the National Research, Development and Innovation Office (NRDI Office) released the country's first national Position Paper on Open Science. This document outlines the principles of open science and encourages collaboration between public institutions and private entities to enhance transparency and accessibility in scientific research.
nkfih.gov.hu

The Position Paper emphasizes the importance of open access to scientific publications and data, advocating for practices such as transformative agreements with publishers and self-archiving. It also highlights the need for robust research data management and adherence to FAIR (Findable, Accessible, Interoperable, and Reusable) principles. To support these initiatives, the NRDI Office plans to implement motivational measures, including integrating open science practices into funding calls, thereby promoting collaboration between public and private sectors.

At the European level, the Horizon Europe program is preparing to launch new public-private partnerships starting in 2025. While specific details are yet to be finalized, these partnerships are expected to cover areas such as advanced materials and research infrastructures. Hungary's active participation in Horizon Europe positions it well to engage in these upcoming collaborations, further strengthening its commitment to equitable public-private partnerships in open science.

In summary, Hungary has undertaken significant efforts to promote equitable public-private partnerships in open science, as evidenced by its national Position Paper and proactive measures by the NRDI Office. With upcoming opportunities in European programs, Hungary is poised to continue advancing in this area through 2025 and beyond.

2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy *(ref.: (ii) 17.j, 23.c)*

YES / UNDER DEVELOPMENT / **NO**

Hungary do not has a standalone national monitoring framework exclusively for open science, it integrates open science indicators within its broader STI policy evaluation mechanisms and actively engages in international efforts to monitor and promote open science practices.

However, Hungary has made significant strides in promoting open science, notably through the National Research, Development and Innovation Office's (NRDI Office) Position Paper on Open Science, released in October 2021. This document outlines the principles for developing an open science ecosystem in Hungary.
https://nkfih.gov.hu/openscience/position-paper-on-open-science

While this Position Paper establishes a foundational framework, it does not specify a dedicated national monitoring system or detailed indicators exclusively for open science. However, Hungary actively participates in international initiatives that monitor and evaluate science, technology, and innovation (STI) policies, which encompass aspects of open science. For instance, the Organisation for Economic Co-operation and Development (OECD) provides a set of Main Science and Technology Indicators (MSTI) that track key trends in STI performance across countries, including Hungary.

Hungary is involved in the European Open Science Cloud (EOSC) initiative, aiming to enhance open science practices across Europe. Through this collaboration, Hungary contributes to and benefits from the development of monitoring frameworks and indicators that assess the implementation and impact of open science policies.

If yes or under development, please provide details and links as relevant.

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of *publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)*

The country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure is 1,39%. It is the latest official data of the Hungarian Central Statistical Office from 2023.

3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. *(ref.: (iii) 18.b)*

Number of fixed broadband internet subscriptions per 1000 inhabitants is 372 pcs. The number of fixed line internet subscriptions surpassed in the 3rd quarter of 2024 by 2.1% the one year earlier level, due to the 8.1% increase of the optical segment. The number of SIM cards for mobile internet grew by 7.4% year-on-year, first of all due to the spread of M2M (machine to machine communication) cards. Mobile data traffic grew in one year by close to one-fourth. On average 35 Gbyte data traffic came by one internet access o-ering mobile subscription. Fixed line internet download traffic increased by 3.9% compared to the same period of the previous year.
(According to the latest official data of the Hungarian Central Statistical Office).

3.3 Are there national research and education networks (NRENs)[viii] active in the country? *(ref.: (iii) 18.c)*

YES / NO

If yes, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

HUN-REN Hungarian Research Network
In 2019, the Government established the Eötvös Loránd Research Network (ELKH) as an independent budgetary institution for the research institutes that were formerly attached to the Hungarian Academy of Sciences with the aim of "ensuring the distribution of available resources between the various disciplines and research institutes in a uniform structure, thereby promoting the optimisation of the efficiency of the Hungarian research system, innovation and research development" (Act LXVIII of 2019).

In 2023 the Hungarian Government decided to renewing the Eötvös Loránd Research Network (ELKH) from a results-oriented perspective while preserving the multidisciplinary character of the network. In order to significantly improve the scientific performance of the ELKH network of research institutes and to put it on an international footing, 3 priority interventions need to be implemented:

1) Renewing the governance of the ELKH (renewing the Governing Board, delegating the powers for creating advisory bodies to the Governing Board and the Chair);
2) Complementing the funding logic of the ELKH with a results-based funding pillar, as in the case of the model-changing universities;
3) Putting the ELKH on an international footing by attracting world-class research teams to the Hungarian innovation ecosystem in the coming years, which can significantly raise the quality of domestic research.

Therefore, the ELKH network continued its work under the name “Hungarian Research Network” from 1 September 2023. The aim of the renaming is to ensure that the independent Hungarian research network is easily and clearly identifiable by name among members of both the domestic and international RDI ecosystem. The name of the Hungarian Research Network (HUN-REN) supports the effective communication of Hungarian research results, thus contributing to the international visibility and recognition of the research network and Hungary.

HUN-REN is a leading research-performing organisation, recognised for its commitment to mission-oriented research and excellence. HUN-REN is a key element of Hungary’s research and innovation ecosystem and serves as the main driving force behind it, collaborating closely with government bodies involved in science policy, universities and major industrial stakeholders.

HUN-REN mission
- to acknowledge, foster and protect scientific freedom and promote research ethics and research integrity;

- to strive for scientific excellence in basic and applied research in mathematics and natural sciences, life sciences, social sciences and the humanities;
- to efficiently operate the publicly funded research network with diversity, inclusivity and by equal opportunity;
- to be at the forefront of research and innovation that generates impact in key strategic and global sustainable areas;
- to develop innovative and targeted research programs and policy;
- to foster and advance partnerships internationally and regionally;
- to strengthen synergies between education and research;
- to evolve as a progressive, open and inspiring organisation

HUN-REN Hungarian Research Network operates under the mandate of the National Assembly of Hungary as an independent entity. HUN-REN encompasses 11 research centres, 8 research institutes, and 116 supported research groups conducting basic and applied research across all scientific disciplines, including SHAPE, STEM and life sciences. The Chairman and members of HUN-REN's Governing Board are primarily acclaimed representatives of the research and innovation ecosystem, including academicians.

In 2024 the HUN-REN Headquarters was established which is responsible for managing the research network's operations and supporting research activities. The HUN-REN Headquarters (HUN-REN HQ) is an independent public budgetary institution in Hungary. It was established by the Parliament effective 1 August 2019 with the aim to manage and operate the publicly funded independent research network in Hungary, which constitutes a central pillar of the country's scientific domain.

In September 2024 the Bay Zoltán Applied Research Nonprofit Ltd. joined HUN-REN. As a cutting-edge research organization, its activities are characterized by a commitment to mission-driven research and excellence. Through its targeted, comprehensive research, it contributes to the well-being of society, the solution of global challenges and the improvement of the quality of life. HUN-REN is a key player and main knowledge engine of the Hungarian research and innovation ecosystem, in close cooperation with government bodies involved in science policy, universities and major industrial players.

In Decree No. 1447/2024 (XII.30.), published on 30 December 2024, the government outlined measures for establishing and funding the HUN-REN Hungarian Research Network. Strengthening Hungary's scientific potential is now a key goal enshrined in a dedicated law. This effort is backed by substantial financial resources and an ongoing transformational program focused on organisational and operational development.

Under the government decree, HUF 1 billion will be allocated to fund the initial assets of HUN-REN, while the Government has set aside an additional HUF 2 billion in the 2025 budget of the Ministry of Culture and Innovation to cover tasks related to legal succession.

In its Decree No. 1447/2024 (XII.30.), the Government has authorised the Minister of Culture and Innovation to sign a public task financing agreement with the Hungarian Research Network, establishing the basis for HUN-REN's new performance-led funding system. The agreement will outline the public tasks and services HUN-REN is responsible for, the related performance indicators, and the amount of state funding provided.

The annual accounting for the state funding will be based on the agreement between the state and HUN-REN. This funding will see a significant increase starting this year, with the Government providing an additional HUF 18 billion. By 2026, total budgetary support will reach HUF 84.5 billion, and from 2027 onward, HUN-REN will receive HUF 97 billion annually for its operations.

The government decree clearly shows that the Government views the renewal of the HUN-REN Hungarian Research Network as a strategic priority and is providing long-term funding to support the transformation of Hungarian research.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to establish them.

3.4 Are there national or institutional open science infrastructures[ix] in your country? *(ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)*

YES / UNDER DEVELOPMENT / NO

If yes or under development, please indicate which type(s) of infrastructure:

- federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage.

Hungarian Open Repositories (HUNOR)

The consortium was established by Hungarian higher education institutions and the Library of the Hungarian Academy of Sciences with the aim of developing the practice of open access (OA) in Hungary. The members of HUNOR wish to improve the domestic and international recognition of Hungarian science through the effective dissemination of scientific results, namely by establishing a national infrastructural network of open access repositories, establishing a methodological centre, applying foreign know-how and international standards in Hungary, establishing complementary channels of scientific communication, and building an international network of contacts.

- community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society

National Data Strategy for Open Science (2021)
Hungary has developed the **National Data Strategy for Open Science (2021)**, which focuses on enhancing the availability of research data and ensuring that data produced through publicly funded research is accessible and reusable. This strategy aligns with the EU's open data agenda and focuses on creating a national infrastructure for open data, including open access digital repositories.
The strategy also addresses issues such as data quality, metadata standards, and long-term data preservation, with a focus on enhancing collaboration among national research institutions and ensuring that data is freely accessible to all stakeholders. OpenScience_allasfoglalas.pdf

Hungarian National Data Archive (MTA SZTAKI)

- The HUN-REN Institute for Computer Science and Control (HUN-REN SZTAKI) manages the Hungarian National Data Archive (HDA), which is an open access digital repository for research data. The HDA is part of Hungary's effort to provide a platform for storing, sharing, and disseminating scientific data, including datasets generated by publicly funded research.
- The HDA ensures that scientific data is accessible, reusable, and preserved for future research. It encourages researchers to deposit their data in the repository, following open data principles to make the research process more transparent and reproducible.

- platforms for exchanges and co-creation of knowledge between scientists and society

As part of the Open Science initiative, research is performed in collaboration, and all research data and information generated during the research process is free to use, view

and access. HUN-REN has undertaken a leading role in supporting Open Science in Hungary. This includes: HUN-REN HQ has signed the National Position Paper on Open Science, a statement initiated by the Hungarian National Research, Development and Innovation Office which describes the principles of open science that best serve the interests and development of Hungarian science, and also summarizes its main areas of activity.
https://hun-ren.hu/open_science

HUN-REN CLOUD

HUN-REN HQ has contributed to the significant upgrading of the capacity of the HUN-REN Cloud, one of Hungary's most important IT cloud systems for research purposes, and also supports the HUN-REN Cloud service hosted by the HUN-REN Institute for Computer Science and Control and the HUN-REN Wigner Research Centre for Physics.

The main objectives of the HUN-REN Cloud are:

- To provide a computing infrastructure of European standard and capacity to HUN-REN researchers and the entire Hungarian research community.
- To provide support for artificial intelligence research.
- To promote a culture of cloud use among Hungarian researchers.
- To support researchers in adapting their applications to the cloud.
- To connect to the European IT infrastructure development ecosystem.

The HUN-REN Cloud has joined the European Open Science Cloud (EOSC) initiative, which provides an open, multidisciplinary IT environment for researchers, developers, businesses and citizens.
The HUN-REN Cloud is part of the latest roadmap of the European Strategy Forum on Research Infrastructures (ESFRI). As part of the SLICES initiative, the roadmap outlines a 10-15-year timeframe for providing a pan-European pilot infrastructure for digital sciences in order to develop the internet of the future.
https://science-cloud.hu/en/hun-ren-cloud

HUN-REN Data Repository Platform (HUN-REN ARP)
The HUN-REN ARP is a new repository infrastructure service that supports the continuous and long-term research data management of the entire HUN-REN research network. The development provides a solution for the secure digital storage, use and reuse of research data as a valuable data asset, as well as its further utilization by the research community and other sectors like the economy.
The ARP Data Repository is not just a repository, but a paradigm shift initiative in the way scientific data is approached, organized, and shared. With its sophisticated features, it is designed to promote professional data management techniques, ensuring the integrity, accessibility, and longevity of research data.

The main characteristics of the HUN-REN Data Repository include:

- The new platform supports the increased visibility, accessibility and long-term conservation of Hungarian scientific findings.
- Three HUN-REN research sites, the HUN-REN Institute for Computer Science and Control, the HUN-REN Centre for Social Sciences and the HUN-REN Wigner Research Centre for Physics are involved in the implementation of the project.
- As part of the open access concept, HUN-REN HQ and the HUN-REN research sites, as member institutions of the Electronic Information Service National Programme (EISZ) have the option for open access publication in any scientific journals where the EISZ has an agreement with the publishers. In addition, users can also access these journals either free of charge or at a discount.

The new Data Repository Platform (ARP) I greatly contributes to the achievement of the objectives set out in Hungary's Research, Development, and Innovation Strategy (2021-2030), which include the increased utilization of research results from public research institutions as well as the expansion of the capacity of knowledge flow and production.

HUN-REN future's mission: key tasks is to disseminate knowledge about the use of data repositories, to establish the necessary data management policies and regulations both at the HUN-REN and at the institutional level, and to introduce domestic and international data and metadata management and storage standards and recommendations, thus enabling the establishment of FAIR (findable, accessible, interoperable, reusable) data repository culture within HUN-REN's institutional network.

The creation of a state-of-the-art domestic research data storage infrastructure will facilitate Hungary's involvement in European initiatives with similar objectives such as the pan-European research infrastructure EOSC (European Open Science Cloud) at an institutional, technological, and direct data link level.
The ARP Data Repository hosts research data provided by the HUN-REN research network and their partners at Hungarian higher education institutions. However, the research data hosted in the Data Repository will be accessible to a wider audience, including the national and international research community, higher education and the competitive sector, as well as the general public interested in science.
The new data repository service was made available to users from the beginning of March 2024. On 25 November 2024 the data Repository was officially opened for HUN-REN researchers and all interested parties.
https://hun-ren.hu/open_science/hun-ren-data-repository-107544

- **Research data and data stewards in Hungary** who iare the link between research data and data repositories :
    - HUN-REN ARP; HUN-REN Wigner Research Centre for Physics)
    - ELI-ALPS Research Institute
    - Corvinus University Budapest, University Library
    - ELTE University Library and Archives
    - National Liboratory for Digital Heritage
    - HUN-REN Wigner Research Centre for Physics, Library; HUN-REN ARP

https://researchdata.hu/en/our-activities

- community-based monitoring and information systems to complement national, regional and global data and information systems.

  HUN-REN Data Repository Platform (HUN-REN ARP)

  The HUN-REN ARP is a new repository infrastructure service that supports the continuous and long-term research data management of the entire HUN-REN research network. The development provides a solution for the secure digital storage, use and reuse of research data as a valuable data asset, as well as its further utilization by the research community and other sectors like the economy.
  The ARP Data Repository is not just a repository, but a paradigm shift initiative in the way scientific data is approached, organized, and shared. With its sophisticated features, it is designed to promote professional data management techniques, ensuring the integrity, accessibility, and longevity of research data.

  The main characteristics of the HUN-REN Data Repository include:
    - The new platform supports the increased visibility, accessibility and long-term conservation of Hungarian scientific findings.

- Three HUN-REN research sites, the HUN-REN Institute for Computer Science and Control, the HUN-REN Centre for Social Sciences and the HUN-REN Wigner Research Centre for Physics are involved in the implementation of the project.
- As part of the open access concept, HUN-REN HQ and the HUN-REN research sites, as member institutions of the Electronic Information Service National Programme (EISZ) have the option for open access publication in any scientific journals where the EISZ has an agreement with the publishers. In addition, users can also access these journals either free of charge or at a discount.
- The new Data Repository Platform (ARP) I greatly contributes to the achievement of the objectives set out in Hungary's Research, Development, and Innovation Strategy (2021-2030), which include the increased utilization of research results from public research institutions as well as the expansion of the capacity of knowledge flow and production.

https://hun-ren.hu/open_science/hun-ren-data-repository-107544

For each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? *(ref.: (iii) 18.e, 18.f)*

YES / NO

Hungary is a member of the European Open Science Cloud (EOSC) Association, contributing to the development of a federated and open multi-disciplinary environment for research data in Europe. The country has a network of repositories, high-speed telecommunication networks, and Authentication and Authorization Infrastructure (AAI) capable of advancing open science. The national initiative, led by the Governmental Agency for IT Development (KIFÜ) and the University of Debrecen, prepares stakeholders for participation in the EOSC ecosystem by increasing awareness and involving local entities in related activities.

Hungary's participation in the NI4OS-Europe project, part of the European Open Science Cloud (EOSC) initiative, has focused on promoting open science in Central and Eastern Europe. Efforts include organizing events and creating resources to engage researchers and librarians in open science practices.

The Hungarian Research Network (HUN-REN) has significantly upgraded the capacity of the HUN-REN Cloud, one of Hungary's most important IT cloud systems for research purposes. This infrastructure provides European-standard computing resources to Hungarian researchers and supports artificial intelligence research, promoting a culture of cloud use among the research community.
https://hun-ren.hu/open_science/hun-ren-cloud-107543

HUN-REN Data Repository Platform
The creation of a state-of-the-art domestic research data storage infrastructure will facilitate Hungary’s involvement in European initiatives with similar objectives such as the pan-European research infrastructure EOSC (European Open Science Cloud) at an institutional, technological, and direct data link level.
https://hun-ren.hu/open_science/hun-ren-data-repository-107544

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open sciencei, stage of research, accessibility to all the relevant users in the region or internationally).

If no, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? *(ref.: (iii) 18.g)*

YES / **NO**

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open sciencei, stage of research, accessibility to all the relevant users in the region or internationally).

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? *(ref.: (iv) 19.a, 19.c)*

YES / NO

Corvinus University of Budapest collects and shares all publications, PhD dissertations and theses produced by CUB students and staff in three repositories. All repositories are maintained by the University Library. Corvinus Research, Corvinus Ph.D. Dissertations, Corvinus Theses, Digital Library of the History of Hungarian Academic Thought
https://www.uni-corvinus.hu/main-page/research/repositories/?lang=en

ELTE University Library and Archives - ELTE Digital Institutional Repository (EDIT)!

https://edit.elte.hu/xmlui/?locale-attribute=en
https://eltekonyvtarak.elte.hu/en/egyetemi-konyvtari-szolgalat/adatbazisok/adatbazislista

Central Library of Semmelweis University

The Central Library of Semmelweis University support the Open Access publishing of articles that's authors (researchers, PhD students) has affiliation with Semmelweis University. The University has a Repository.
https://lib.semmelweis.hu/open_access_support

If yes, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

YES / NO

If yes, please provide more information and links as relevant.

AI Ambassadors Programm of the Hungarian Research Network (HUN-REN) connect researchers with AI technology opportunities.
The primary objective of HUN-REN is to support research effectiveness, and an important tool for this will be the ability to use artificial intelligence. The development of this capability is at the heart of the HUN-REN Artificial Intelligence Action Plan.

The AI Action Plan introduced in 2024 is aimed at helping all the HUN-REN network's institutes and research centres adopt the best AI technologies, thus contributing to further improving their scientific performance.

https://hun-ren.hu/hunren_news/roland-jakab-ai-ambassadors-connect-researchers-with-ai-technology-opportunities-106408

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? *(ref.: (iv) 19.b)*

YES / **NO**

The broader educational landscape in Hungary is evolving to include competencies that align with open science principles. For instance, the 2020 national core curriculum emphasizes sustainability and environmental awareness as key educational goals, reflecting an interdisciplinary approach that resonates with open science values.

Hungary's Digital Education Strategy, launched in 2016, addresses all levels of education, including higher education, and emphasizes the development of digital competencies. Open Source solutions play a significant role in this strategy, providing tools and educational materials that support both educators and students.

These initiatives indicate progress towards integrating open science competencies into higher education, a fully established framework within research skills curricula is still be under development. Given the ongoing efforts and the strategic emphasis on open science, it is plausible to anticipate that such a framework could be implemented by the end of 2025.

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

4.4 Are open educational resources [x] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? *(ref.: (iv) 19.d)*

YES/ PARTLY/ NO

Open Science Lecture Series in Hungarian Higher Education
The Open Science Lecture Series in Hungarian Higher Education was held 22 times at 8 venues between 22 September and 9 December 2022. The topics of the lecture series included introduction to Open Science, the concept of open access, research data management, research evaluation systems, participants were introduced to the services of the European Open Science Cooperation, and those who attended a workshop could gain knowledge about publications. The Government Information Technology Development Agency reached more than 370 students with a series of lectures.
https://openscience.hu/events/magyar-nyilt-tudomanyos-forum-vi/

4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? *(ref.: (iv) 19.e)*

YES / NO

Many initiatives reflect Hungary's commitment to fostering open science practices, enhancing interdisciplinary collaboration, and making scientific knowledge accessible to researchers, policymakers, and the public.

In October 2021, the National Research, Development and Innovation Office (NRDI Office) launched Hungary's first national Position Paper on Open Science. This document, developed by the Open Science Advisory Board established in March 2021, outlines the principles for developing an open science ecosystem in the country. It emphasizes transparency and collaboration, aiming to make research, development, and innovation results accessible to all. The NRDI Office plans to implement motivational measures to encourage adoption of open science practices, including integrating open science criteria into funding calls.

The Hungarian Research Network (HUN-REN) has also taken a leading role in supporting open science. By promoting collaborative research and ensuring that all research data and information generated are freely accessible, HUN-REN enhances the dissemination of Hungarian research achievements, thereby increasing their international visibility and recognition.

Hungary has been instrumental in global open access movements. The Budapest Open Access Initiative, launched in 2002, is a significant milestone advocating for free and unrestricted access to scholarly literature. This initiative has influenced open access policies worldwide, promoting the dissemination of scientific knowledge to a broad audience.

The World Science Forum, an international conference series on global science policy, has been organized biennially in Budapest since 2003. This forum serves as a platform for dialogue between the scientific community and society, addressing the role and responsibility of science in the 21st century.

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? *(ref.: (v) 20.b, 20c, 20i)*

YES / NO

HUN-REN HQ has launched the HUN-REN Data Repository project, which aims to develop a digital infrastructure for the management of research data according to FAIR (Findable, Accessible, Interoperable and Reusable) principles.
https://hun-ren.hu/open_science/hun-ren-data-repository-107544

HUN-REN launched even the Hungrian Open Science Cloud project.
https://science-cloud.hu/en/hun-ren-cloud

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? *(ref.: (v) 20.a, 20.c, 20.d)*

YES / **NO**

Hungary has embraced open science at a policy level, the integration of specific clauses recognizing open science practices into researcher career evaluations and progression systems remains an area for potential development. Aligning national policies with European frameworks could further promote the adoption of open science practices within Hungary's research community.

The National Research, Development and Innovation Office (NRDI Office) released the first national-level Position Paper on Open Science to guide this effort. While this position paper signifies a strategic move towards open science, specific details regarding the incorporation of open science practices—such as sharing, collaboration, and engagement with societal actors—into Hungary's career evaluation and progression systems for researchers are not explicitly outlined in the available sources.
The list of criteria for selecting reviewers needs to be publicished. The work of reviewers should be duly recognised, both in terms of the reviewers' career-assessment and in terms of fair remuneration. The transparency of the assessment process, the publication of the evaluation guidelines and criteria, as well as the assessment forms and the minutes of jury meetings will all increase confidence in the application system, while at the same time provide important lessons for applicants to better understand the evaluation system and improve their future applications, should their application not be approved.

At the European level, the European Charter for Researchers and the Code of Conduct for the Recruitment of Researchers provide a framework promoting open science practices. These documents advocate for the recognition of activities like public engagement and dissemination of results in researcher evaluations.
The European Commission's report "Evaluation of Research Careers Fully Acknowledging Open Science Practices" emphasizes the need for comprehensive assessment criteria that include open science activities. It recommends that research institutions and funding agencies adapt their evaluation systems to recognize and reward open science contributions.

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? *(ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)*

YES / NO

The National Research, Development and Innovation Office (NRDI Office) released national-level Position Paper on Open Science to guide the effort on new types of rewards and initiatives.

The future performance and competitiveness of a new Hungarian research, development and innovation ecosystem largely depend on the closer cooperation between stakeholders, such as policy-makers, higher education and research institutions, businesses and professional

organisations. Endorsing the concept of Open Science throughout the national innovation ecosystems promotes transparency and interoperability, and the sharing of best practices and results can inspire further research and innovation projects.
The Signatories support the inclusion of the concept of Open Science as an important aspect in the frameworks of domestic research funding organisations, supporting both individual research excellence and research at the institutional level (e.g., thematic programmes). Thus, the dissemination of a new perspective can be greatly enhanced, and the networks of National Laboratories and Territorial Innovation Platforms can apply this new approach in RDI projects, networking and social communication.

If yes, for each initiative, please provide more information, including on the stakeholders[xi] that are involved or lead the initiative, and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

5.4 Have there been any unintended negative consequences[xii] of open science practices reported in your country? (ref.: (v) 20)

YES / NO

In October 2023, the Hungarian Academy of Sciences (MTA) prepared a report entitled *"Proposals for the management of articles in journals engaging in objectionable practices – MTA's recommendations in relation to new types of publication misuse*", which was published on their website and in print in January 2024.

- At its meeting on 18 April 2023, the Presidium of the Hungarian Academy of Sciences decided to set up a committee whose main task is to develop an action plan against predatory communication practices for the Academy and the practitioners of science.
- The committee analysed the causes of abuses and the steps that need to be taken by the research community and organisations responsible for science policy to preserve the credibility of science and strengthen trust in science.
- The committee's work resulted in a report entitled "Proposals for the management of articles in journals with objectionable practices". published on the website of the Hungarian Academy of Sciences.

If yes, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? *(ref.: (vi) 21)*

YES, at the national level.

YES, at the institutional level.

YES, at both national and institutional levels.

National Declaration on Open Science - Invitation by National Research, Development and Innovation Office
https://nkfih.gov.hu/nyilt-tudomany

Position paper on Open Science of National Research, Development and Innovation Office (NRDI Office)

Next generation metrics in research assessment

The spread of Open Science necessitates a revision of the practice of benchmarking based on quantitative methodologies, by introducing a new, holistic approach that considers the research assessment processes as a fundamental quality assurance condition for the creation of new knowledge and innovation, to support social and economic development. The evaluation should primarily focus on the content of the research.

- Methods for evaluating scientific performance should support evaluators in carrying out qualitative assessments that take into account a broad spectrum of research outputs and activities.
- Research evaluation processes should be clear and transparent at all stages and for all stakeholders. Research institutions should continuously monitor and regularly review the reliability of their own evaluation processes. The updated guidelines should be accessible to all stakeholders.
- Research funding organisations should streamline evaluation processes to ensure transparency and reduce the burden on evaluators and applicants. The standardisation and interoperability of the whole application process within organisations and in line with other national and international organisations would be advantageous.
- Organisations need to unambiguously define the meaning of any terms they use to describe research performance (e.g., excellence), clearly highlighting where evaluation methods use different, discipline- or career stage-specific criteria to define quality. Comparing assessments for research purposes will help improve the operational efficiency of these processes.

The Signatories recommend that all research assessment organisations publish freely accessible and user-friendly guidelines on all the processes they follow when conducting assessments. The guidelines should include a description of the criteria and methodology used for the assessment. However, research funding organisations and decision-making bodies should consider incorporating a response-and-feedback mechanism into their assessment processes to improve the quality of assessments.
https://nkfih.gov.hu/openscience/position-paper-on-open-science

NO

If yes, please indicate which of the recommended actions they address and provide more information and links as relevant:

- promotion of pre-prints
- exploration of open peer-review practices
- valuing and sharing of negative scientific results and associated data
- participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas.
- promoting open innovation practices.
- Others (with a box to add text)

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? *(ref.: (vii) 22.a)*

Position paper on Open Science of National Research, Development and Innovation Office (NRDI Office)
International cooperation
The European Open Science Cloud (*EOSC*) is an ecosystem for the storage and processing of research data and the development of a common European research infrastructure platform in support of European science. The European Commission set up this initiative with the aim of promoting a more coherent and compatible operation of existing European research data infrastructures and to achieve an alliance of data and related infrastructures, in accordance with FAIR-principles, among others.
Launched in 2015 and supported by the European Commission, the EOSC initiative is a European partnership, planned and funded under *Horizon Europe* as of 2021. Established in July 2020, the EOSC *Association* is a unified and formal organization with a priority to dynamically expand its membership.
The Signatories encourage Hungarian scientific-, research-, development- and innovation organisations and institutions to join EOSC, thus promoting international collaboration and the visibility of Hungarian science. In addition, the Signatories support international co-operation that is capable of improving the position of Hungarian science globally, promoting the completion of researchers' careers, and fostering multidisciplinary and intersectoral research.
https://nkfih.gov.hu/openscience/position-paper-on-open-science

World Science Forum - Focus on Open Science
Organised by: Library and Information Centre of the Hungarian Academy of Sciences (MTA KIK), Association of European Research Libraries (LIBER), Scientific Knowledge Services (SKS)
The Library and Information Centre of the Hungarian Academy of Sciences, in collaboration with Scientific Knowledge Services and LIBER (Association of European Research Libraries), organises the annual Focus on Open Science one-day international conference on research support as a satellite event of World Science Forum.
The conference aimed to bring together research support professionals from the Central European region, to provide an opportunity to exchange experiences with the most eminent European representatives of the field, to promote the concept of open science and scholarly communication, with a special focus on cooperation, tasks and opportunities for researchers and libraries.The Library and Information Centre of the Hungarian Academy of Sciences, in collaboration with Scientific Knowledge Services and LIBER (Association of European Research Libraries), organises the annual Focus on Open Science one-day international conference on research support as a satellite event of World Science Forum.
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Hungarian Implementation of the EU Open Data Directive

- Hungary is required to comply with the EU Open Data Directive (2019/1024), which mandates the open publication of government data and encourages public institutions to make research and scientific data publicly accessible. This Directive builds on the earlier Public Sector Information Directive (2003/98/EC), which introduced the legal framework for making public sector data available in reusable formats.

- The Hungarian government has adapted its policies and laws to meet these requirements, ensuring that publicly funded research data and publications are accessible through national and institutional open access repositories.

European Open Science Cloud (EOSC) Participation

- The EOSC facilitates cross-border data sharing and promotes the use of open data repositories to make scientific research more transparent and collaborative. It is very important for the enhanced capacity HUN-REN Cloud to integrate into the European EOSC (European Open Science Cloud) program and the ESFRI infrastructure programs. To this end, HUN-REN SZTAKI is already participating in two European projects. In the EGI-ACE project, we ensure the compatibility of the HUN-REN Cloud with the EOSC clouds and provide user support for European projects where Hungarian consortium members are involved. In the SLICES-SC project, we are contributing to the preparation of the SLICES ESFRI program and organizing the European use of the SLICES infrastructure. Our goal is to participate in as many European projects as possible in the future, catering to the cloud requirements of the respective projects
- Through its participation in EOSC, Hungary is contributing to the development of national and international infrastructures for data sharing, and Hungarian research institutions and researchers can take part in open science practices facilitated by EOSC. https://hun-ren.hu/open_science/hun-ren-cloud-107543

7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? *(ref.: (vii) 22.b, 22.e, 22.f)*

YES / NO

National Declaration on Open Science - Invitation by National Research, Development and Innovation Office

The EOSC initiative, supported by the European Commission and launched in 2015, has been operating as a European partnership planned and funded under Horizon Europe since 2021. The EOSC Association, established in July 2020, implements a unified and formal representation of interests, and its main goal is to expand its membership dynamically.

The signatories of the Declaration encourage Hungarian scientific, research, development and innovation organisations and institutions to join the EOSC, thus promoting international collaboration and the visibility of the results of Hungarian science. In addition, the Signatories support all international cooperation that improves the perception of Hungarian science and supports the completion of research careers

https://nkfih.gov.hu/hivatalrol/strategia-alkotas/open-science/allasfoglalasű

Horizon Europe

EOSC-hub project

The EOSC-hub project creates the integration and management system of the future European Open Science Cloud that delivers a catalogue of services, software and data from the EGI Federation, EUDAT CDI, INDIGO-DataCloud and major research e-infrastructures.

This integration and management system (the Hub) builds on mature processes, policies and tools from the leading European federated e-Infrastructures to cover the whole life-cycle of services, from planning to delivery.

The Hub aggregates services from local, regional and national e-Infrastructures in Europe, Africa, Asia, Canada and South America.

The Hub acts as a single contact point for researchers and innovators to discover, access, use and reuse a broad spectrum of resources for advanced data-driven research. Through the virtual access mechanism, more scientific communities and users have access to services

supporting their scientific discovery and collaboration across disciplinary and geographical boundaries.
The project also improves skills and knowledge among researchers and service operators by delivering specialised trainings and by establishing competence centres to co-create solutions with the users. In the area of engagement with the private sector, the project creates a Joint Digital Innovation Hub that stimulates an ecosystem of industry/SMEs, service providers and researchers to support business pilots, market take-up and commercial boost strategies.
EOSC-hub builds on existing technology already at TRL 8 and addresses the need for interoperability by promoting the adoption of open standards and protocols. By mobilizing e-Infrastructures comprising more than 300 data centres worldwide and 18 pan-European infrastructures, this project is a ground-breaking milestone for the implementation of the European Open Science Cloud.
https://sztaki.hun-ren.hu/en/innovation/projects/eosc-hub

If yes, please provide links as relevant and more information.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=21)

YES / NO

Hungary is actively engaged in discussions and initiatives to establish regional and international funding mechanisms for open science.

Science Patronage Programme
In 2021, the National Research, Development and Innovation Office (NRDI Office) launched the Science Patronage Programme, allocating 1 billion Hungarian Forints (approximately 2.6 million Euros) to promote open science. NRDI Office has awarded funding to 366 researchers and research centres under the Science Patronage Programme call, financed from the National Research, Development and Innovation Fund.
This programme encourages the Hungarian research community's involvement in international scientific platforms and events, fostering the advancement of open science in Hungary. A portion of this budget is specifically dedicated to supporting open access publications.

In 2024 aimed at encouraging the integration of the Hungarian scientific research community into the international arena, supporting science communication, and promoting the expansion of open science in Hungary, the call published on 24 April 2024, with a funding budget of HUF 1 billion, comprised three sub-programs:

- Participation in a conference abroad
- Organisation of an international conference in Hungary
- Promotion of scientific results

A total of 452 valid proposals were received by the deadline, with a total funding request of HUF 5,238,661,000. In consideration of the significant number and high professional quality of the submitted proposals, and in accordance with the decision of the Ministry for Culture and Innovation, the call provides increased funding of nearly 2.5 billion HUF compared to the original call.
This boost aims to enhance the visibility of domestic scientific results, facilitate the internationalisation of the Hungarian research community, and strengthen openness toward scientific and innovation outcomes.
https://nkfih.gov.hu/english/nrdi-fund/funded-projects-mec24

Collaboration with Austrian Science Fund (FWF)
Hungary collaborates with Austria through a joint funding initiative between the NRDI Office and the Austrian Science Fund (FWF). This programme supports closely integrated research projects between Austrian and Hungarian researchers across all disciplines, with the current

European Istitute of Innovation and Techmology
Hungary is a member of the European Institute of Innovation and Technology (EIT), an independent EU body established to strengthen Europe's innovation capacity. The EIT supports entrepreneurial education, business creation, and innovation-driven research projects, contributing to the broader European open science agenda.

If yes, please provide links as relevant and more information.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

8. GENERAL CONSIDERATIONS

8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

YES/**NO**

If yes, please provide more information.

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i Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

ii The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.
Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.
Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

iii National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

iv These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

v Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

vi A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

vii Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

viii A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

ix Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

x Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

xi For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.

xii For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.